NATIONAL MATHEMATICS CONTEST
A-LEVEL PAPER 1 QUESTIONS AND SOLUTIONS.

SECTION A: 5 marks each

1. Find the real-valued function $f(x)$ such that $f\left(x-\frac{1}{x}\right)=x^3-\frac{1}{x^3}$.

Solution:

Using the fact that

$$\left(x-\frac{1}{x}\right)^3=x^3-3x^2\cdot\frac{1}{x}+3x\cdot\frac{1}{x^2}-\frac{1}{x^3}=x^3-\frac{1}{x^3}-3\left(x-\frac{1}{x}\right),$$

we get that

$$x^3-\frac{1}{x^3}=\left(x-\frac{1}{x}\right)^3+3\left(x-\frac{1}{x}\right).$$

Thus,

$$f\left(x-\frac{1}{x}\right)=x^3-\frac{1}{x^3}=\left(x-\frac{1}{x}\right)^3+3\left(x-\frac{1}{x}\right),$$

which implies that $\boxed{f(x)=x^3+3x}$ is the required function.

2. Show that the infinite tower, $\sqrt{2}^{\sqrt{2}^{\sqrt{2}^{\cdot^{\cdot^{\cdot}}}}}=2$.

[**Hint**: use the property of the **Lambert** W function that $W\left(\psi e^{\psi}\right)=\psi$ for any function ψ]

Solution:

Let $\sqrt{2}^{\sqrt{2}^{\sqrt{2}^{\cdot^{\cdot^{\cdot}}}}}=x$. So, $\sqrt{2}^x=x$. Taking the natural logarithm on both sides, we get $\ln\sqrt{2}^x=\ln x$, which is equivalent to $x\ln\sqrt{2}=\ln x$. This is equivalent to $e^{\ln x}\ln\sqrt{2}=\ln x$. Thus, $\ln\sqrt{2}=\ln xe^{-\ln x}$. Multiplying through both sides by -1, we get that

$$-\ln\sqrt{2}=-\ln xe^{-\ln x}.$$

But $-\ln\sqrt{2}=-\ln 2^{\frac{1}{2}}=-\frac{1}{2}\ln 2=\frac{1}{2}\ln 2^{-1}=\frac{1}{2}\ln\frac{1}{2}=\ln\frac{1}{2}e^{\ln\frac{1}{2}}$. Therefore, we get that

$$\ln\frac{1}{2}e^{\ln\frac{1}{2}}=-\ln xe^{-\ln x}=\ln x^{-1}e^{\ln x^{-1}}=\ln\frac{1}{x}e^{\ln\frac{1}{x}}.$$

Now, applying the Lambert W function to both sides of the above equation, we get that

$$\ln\frac{1}{2}=W\left(\ln\frac{1}{2}e^{\ln\frac{1}{2}}\right)=W\left(\ln\frac{1}{x}e^{\ln\frac{1}{x}}\right)=\ln\frac{1}{x}.$$

This gives that $\ln\frac{1}{2}=\ln\frac{1}{x}$, or $e^{\ln\frac{1}{2}}=e^{\ln\frac{1}{x}}$, or $\frac{1}{2}=\frac{1}{x}$, which implies that $\boxed{x=2}$, as required.

3. Given that $f\left(\dfrac{x+2}{x-2}\right) = \dfrac{x^2+4x+4}{8x}$, find the function $f(x)$.

Solution:

First, we observe that

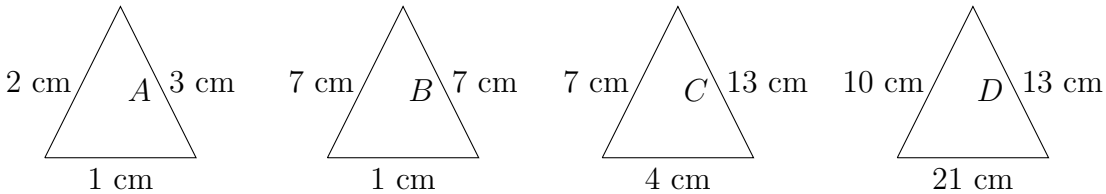
$$f(0) = f\left(\frac{-2+2}{-2-2}\right) = \frac{(-2)^2+4(-2)+4}{8(-2)} = \frac{4-8+4}{-16} = \frac{0}{-16} = 0.$$

Now, let $y = \frac{x+2}{x-2}$. This implies that $y(x-2) = x+2$, or $yx - x = 2y+2$, or $x(y-1) = 2(y+1)$. This implies that $x = 2\left(\frac{y+1}{y-1}\right)$. Thus, we get

$$\begin{aligned}
f(y) &= \frac{\left(2\left(\dfrac{y+1}{y-1}\right)\right)^2 + 4\left(2\left(\dfrac{y+1}{y-1}\right)\right) + 4}{8\left(2\left(\dfrac{y+1}{y-1}\right)\right)} \\
&= \frac{\left(\dfrac{y+1}{y-1}\right)^2 + 2\left(\dfrac{y+1}{y-1}\right) + 1}{4\left(\dfrac{y+1}{y-1}\right)} \\
&= \frac{(y+1)^2 + 2(y+1)(y-1) + (y-1)^2}{4(y+1)(y-1)} \\
&= \frac{y^2+2y+1+2y^2-2+y^2-2y+1}{4(y^2-1)} \\
\therefore f(y) &= \frac{4y^2}{4(y^2-1)} = \frac{y^2}{y^2-1}.
\end{aligned}$$

Finally, we also observe that $f(0) = \dfrac{0^2}{0^2-1} = \dfrac{0}{-1} = 0$. Therefore, $\boxed{f(x) = \dfrac{x^2}{x^2-1}}$.

4. Which of the following triangles A, B, C, D does not exist?



Solution:

We use the fact that if a, b, c are the sides of a triangle, then following conditions $a+b \geq c$, $a+c \geq b$, and $b+c \geq a$, must be satisfied.

- For triangle A, we see that $2+3 = 5 \geq 1$, $1+2 = 3 \geq 3$, and $1+3 = 4 \geq 2$. Thus, triangle A exists.
- For triangle B, we see that $1+7 = 8 \geq 7$, $7+7 = 14 \geq 1$, and $7+1 = 8 \geq 7$. Thus, triangle B also exists.

- For triangle C, we see that $7+13=20\geq 4$, $4+13=17\geq 7$, but $7+4=11<13$. Thus, **triangle C does not exist**.
- For triangle D, we observe that $10+21=31\geq 13$, $21+13=34\geq 10$, and $10+13=23\geq 13$. Thus, triangle D also exists.

5. An unreliable typist can guarantee that when they try to type a word with different letters, every letter of the word will appear exactly once in what they type, and each letter will occur at most one letter late (though it may occur more than one letter early). Thus, when trying to type **MATHS**, the typist may type **MATHS, MTAHS** or **TMASH**, but not **ATMSH**. Determine, with proof, the number of possible spellings of **OLYMPIADS** that might be typed.

Solution:

The number of possible spellings is equivalent to the number of cycles formed from the number of letters. Let the number of different letters in a word to be typed be n.

- For $n=2$, e.g. (●●) two arrangements (●●), (●)(●) are possible. That is, the number of cycles $=2=2^1=2^{2-1}$.
- For $n=3$, e.g. (● ● ●) four arrangements (● ● ●), (●)(●●), (●●)(●), (●)(●)(●) are possible. That is, the number of cycles $=4=2^2=2^{3-1}$.
- For $n=4$, e.g. (● ● ●●) eight possible arrangements (● ● ●●), (●)(● ● ●), (● ● ●)(●), (●●)(●●), (●●)(●)(●), (●)(●●)(●), (●)(●)(●●), (●)(●)(●)(●) are possible. That is, the number of cycles $=8=2^3=2^{4-1}$.
- In general, for a word with n different letters, the number of possible cycles is 2^{n-1}.
- For **OLYMPIADS**, $n=9$. Thus, the number of possible spellings that can be typed is $=2^{9-1}=2^8=256$ spellings.

6. Find all positive integers n such that the integer $n\times 2^n+1$ is a perfect square.

[**Hint**: An integer x is a perfect square if there is an integer y such that $x=y^2$]

Solution:

First, we try to check out the solutions for the first few values of n:

n	$n\times 2^n+1$	Is square?
1	$1\times 2^1+1=3$	No
2	$2\times 2^2+1=9=3^2$	**Yes**
3	$3\times 2^3+1=25=5^2$	**Yes**
4	$4\times 2^4+1=65$	No
5	$5\times 2^5+1=161$	No

Next, we prove that indeed the values $n=2$ and $n=3$ are the only solutions. Let $n\cdot 2^n+1=m^2$. That is, $n\cdot 2^n=m^2-1=(m-1)(m+1)$. Thus, either $m+1=a\cdot 2^{n-1}$ or $m-1=a\cdot 2^{n-1}$.

Case 1: If $m-1=a\cdot 2^{n-1}$, then $n\cdot 2^n=(m+1)a\cdot 2^{n-1}$, which implies that $2n=a(m+1)$. That is, $2n\geq m+1\geq m-1\geq 2^{n-1}$ or $2n\geq 2^{n-1}$ or $n\geq 2^{n-2}$, which is only true if $n\leq 4$.

Case 2: If $m+1=a\cdot 2^{n-1}$, then $n\cdot 2^n=(m-1)a\cdot 2^{n-1}$, which implies that $2n=a(m-1)$. That is, $2n\geq m-1=(m+1)-2\geq 2^{n-1}-2$, or $2n\geq 2^{n-1}-2$, or $n\geq 2^{n-2}-1$, which only holds when $n\leq 4$.

Thus, in both cases the solutions must satisfy the inequality $1\leq n\leq 4$. Thus, $\boxed{n=2,3}$ are the only possible values of n such that $n\times 2^n+1$ is a perfect square.

7. Let x, y, z be real numbers. Prove that if

$$\frac{xy+1}{x+1} = \frac{yz+1}{y+1} = \frac{zx+1}{z+1} \tag{1.1}$$

then $x = y = z$. [**Hint**: Use the **AM-GM Inequality**]

Solution:

From the first equality in (1.1), we have

$$\frac{xy+1}{x+1} = \frac{yz+1}{y+1} \quad \Rightarrow \; (xy+1)(y+1) = (yz+1)(x+1),$$

so that

$$xy^2 + xy + y = xyz + yz + x. \tag{1.2}$$

Similarly, we also have that

$$yz^2 + yz + x = xyz + zx + y, \tag{1.3}$$
$$zx^2 + zx + x = xyz + xy + z. \tag{1.4}$$

Now, adding (1.2), (1.3) and (1.4) gives: $xy^2 + yz^2 + zx^2 = 3xyz$, or

$$\frac{x}{y} + \frac{y}{z} + \frac{z}{x} = 3. \tag{1.5}$$

Applying the **AM-GM Inequality** on the left-hand side of (1.5) gives:

$$\frac{\frac{x}{y} + \frac{y}{z} + \frac{z}{x}}{3} \geq \left(\frac{x}{y} \cdot \frac{y}{z} \cdot \frac{z}{x}\right)^{\frac{1}{3}} = 1 \quad \Rightarrow \quad \frac{x}{y} + \frac{y}{z} + \frac{z}{x} \geq 3.$$

From this, equality is achieved only when

$$\frac{x}{y} = \frac{y}{z} = \frac{z}{x} = 1 \quad \Rightarrow \quad \boxed{x = y = z}.$$

8. Let g be a function with the following properties:

   (i) $g(1) = 1$; and

   (ii) $g(5n) = nf(n)$ for any positive integer n.

   What is the value of $g(5^{100})$?

Solution:

Using the property (ii), we get that

$$\begin{aligned} g(5^{100}) &= g(5 \cdot 5^{99}) = 5^{99} \cdot g(5^{99}) = 5^{99} \cdot g(5 \cdot 5^{98}) = 5^{99} \cdot 5^{98} \cdot g(5^{98}) = 5^{99+98} \cdot g(5^{98}) \\ &= \cdots = 5^{99+98+\cdots+2+1+0} \cdot g(5^0) = 5^{99+98+\cdots+2+1+0} \cdot g(1) = 5^{99+98+\cdots+2+1+0} \cdot 1 \end{aligned}$$

$$\therefore g(5^{100}) = 5^{99+98+\cdots+2+1+0} = 5^{\frac{99 \cdot 100}{2}} = \boxed{5^{4950}}.$$

9. Chess is played on an 8×8 square board. How may ways are there to put 3 identical **rooks** on the chessboard so that no two **rooks** are in the same row or column?

Solution:

Place the rooks in order so that they can not capture, leading to

$$\binom{8}{1}\binom{8}{1} = 64, \quad \binom{7}{1}\binom{7}{1} = 49, \quad \text{and} \quad \binom{6}{1}\binom{6}{1} = 36 \quad \text{possibilities.}$$

Then, divide their product $64 \cdot 49 \cdot 36$ by $3! = 6$ to account for the reorderings. Therefore, the answer is $\dfrac{64 \cdot 49 \cdot 36}{6} = \boxed{18816}$ possible ways.

10. Let ABC be an acute-angled triangle. Point P is such that $AP = AB$ and $PB \| AC$. Point Q is such that $AQ = AC$ and $CQ \| AB$. Line segments CP and BQ meet at point X. Prove that the circumcenter of triangle ABC lies on the circumcircle of triangle PXQ.

Solution:

Let D be the vertex of parallelogram $ABDC$. Then $APDC$ and $AQDB$ are isosceles trapezoids. Therefore the perpendicular bisectors to line segments PD and QD coincide with the perpendicular bisectors to AC and AB respectively, the circumcenter O of triangle ABC is also the circumcenter of DPQ and $\angle POQ = 2\angle BAC$. Also since, $\angle XPD = \angle ADP$, $\angle XQD = \angle ADQ$ we obtain that $\angle PXQ = 2\angle BAC$. Thus, points O, P, Q, X are concyclic.

SECTION B: 10 marks each

11. Find all values of y that satisfy the equation $(y+1)^2+(y+2)^3+(y+3)^4=2$.

Solution:

Let $x=y+2$, so the equation in question becomes $(x-1)^2+x^3+(x+1)^4=2$. But $(x-1)^2=x^2-2x+1$ and $(x+1)^4=x^4+4x^3+6x^2+4x+1$ (by Pascal's triangle). Thus, we get that

$$\begin{aligned} x^2-2x+1+x^3+x^4+4x^3+6x^2+4x+1&=2\\ \Rightarrow x^4+5x^3+7x^2+2x+2&=2\\ \Rightarrow x^4+5x^3+7x^2+2x&=0\\ \Rightarrow x(x^3+5x^2+7x+2)&=0. \end{aligned}$$

Either: $x=0$, which means that $y+2=0$. So, $\boxed{y=-2}$ is a solution.

Or: $x^3+5x^2+7x+2=0$. Let $f(x)=x^3+5x^3+7x+2$, by the **Rational Root Theorem**, possible roots of $f(x)$ are all the divisors of 2, that is, $x=\pm1,\pm2$. Depending on the way $f(x)$ is defined, it is clear that $f(1)\neq0$ and $f(2)\neq0$. So, we try

$$f(-1)=(-1)^3+5(-1)^2+7(-1)+2=-1+5-7+2=-1\neq0$$

and

$$f(-2)=(-2)^3+5(-2)^2+7(-2)+2=-8+20-14+2=0.$$

Thus, $x=-2$ is a root, or $(x+2)$ is a factor. To find the other factors, we do a **Long Divison**:

$$\begin{array}{r} x^2+3x+1\\ \hline x+2\big)\;\; x^3+5x^2+7x+2\\ \underline{-\,x^3-2x^2}\\ 3x^2+7x+2\\ \underline{-\,3x^2-6x}\\ x+2\\ \underline{-\,x-2}\\ 0 \end{array}$$

Thus, $x^3+5x^2+7x+2=(x+2)(x^2+3x+1)$. **Either**: $x+2=0$, this means that $y+2+2=0$ or $y+4=0$. Therefore $\boxed{y=-4}$ is a solution. **Or**: $x^2+3x+1=0$. Here, we use the **Quadratic Formula** to find the remaining solutions. That is, $x=\dfrac{-3\pm\sqrt{3^2-4}}{2}=\dfrac{-3\pm\sqrt{5}}{2}$. This means that $y+2=\dfrac{-3\pm\sqrt{5}}{2}$, or $y=\dfrac{7\pm\sqrt{5}}{2}$. So, $\boxed{y=\dfrac{7+\sqrt{5}}{2}}$ and $\boxed{y=\dfrac{7-\sqrt{5}}{2}}$ are the remaining solutions. All solutions are

$$\boxed{y\in\left\{-4,\;\;-2,\;\;\frac{7-\sqrt{5}}{2},\;\;\frac{7+\sqrt{5}}{2}\right\}}.$$

12. Find all pairs of nonzero integers (x, y) such that the integer $x^2 + y^2$ is a common divisor of the integers $x^5 + y$ and $y^5 + x$.

Solution:

We are given that $x^2 + y^2 \mid x^5 + y$ and $x^2 + y^2 \mid y^5 + x$. This is equivalent to saying that $x^2 + y^2 \mid x(x^5 + y)$ and $x^2 + y^2 \mid y(y^5 + x)$, and hence $x^2 + y^2 \mid x(x^5 + y) + y(y^5 + x)$. That is,

$$x^2 + y^2 \mid x^6 + y^6 + 2xy. \tag{1.6}$$

From the well-known identity for the sum of cubes, $A^3 + B^3 = (A + B)(A^2 - AB + B^2)$, we also have that $x^6 + y^6 = (x^2)^3 + (y^2)^3 = (x^2 + y^2)((x^2)^2 - x^2y^2 + (y^2)^2)$. This implies that

$$x^2 + y^2 \mid x^6 + y^6. \tag{1.7}$$

From (1.6) and (1.7), we get that $x^2+y^2 \mid 2xy$. This implies that $x^2+y^2 \leq 2|xy|$, or $x^2-2|xy|+y^2 \leq 0$, or $(|x| - |y|)^2 \leq 0$, that is, $|x| = |y|$. Therefore, we have two cases:

- **Case 1**: If $x = y$, then $2x^2 \mid x^5 + x$, which implies that $2x^2 \mid x(x^4 + 1)$, or $2x \mid x^4 + 1$. This means that $x \mid 1$. That is, either $x = 1$ or $x = -1$.
- **Case 2**: If $x = -y$, then $2x^2 \mid x^5 - x$, which implies that $2x^2 \mid x(x^4 - 1)$, or $2x \mid x^4 - 1$. This means that $x \mid 1$, that is, either $x = 1$ or $x = -1$.

Hence, the required solutions are $\boxed{(x, y) \in \{(1, 1), (-1, -1), (1, -1), (-1, 1)\}}$.

13. Let a and b be real numbers with $a \neq 0$. Let $P(x) = ax^4 - 4ax^3 + (5a + b)x^2 - 4bx + b$. Show that all roots of $P(x)$ are real and positive if and only if $a = b$.

Solution:

($\Leftarrow$) Suppose that $a = b$, then with the aid of **Pascal's Triangle**, we have that

$$P(x) = ax^4 - 4ax^3 + 6ax^2 - 4ax + a = a(x^4 - 4x^3 + 6x^2 - 4x + 1) = a(x - 1)^4.$$

Thus, $P(x) = 0$ implies that $a(x - 1)^4 = 0$, which implies that, since $a \neq 0$, then $\boxed{x = 1}$ is the only root, which is real and positive.

($\Rightarrow$) Let r_1, r_2, r_3, r_4 be the real and positive roots of the polynomial $P(x)$. Then $P(x) = 0$ implies that $ax^4 - 4ax^3 + (5a + b)x^2 - 4bx + b = 0$. Since, $a \neq 0$, we may divide through by a to get that:

$$x^4 - 4x^3 + \left(5 + \frac{b}{a}\right)x^2 - 4\left(\frac{b}{a}\right)x + \frac{b}{a} = 0.$$

Then, the roots r_1, r_2, r_3, r_4, must satisfy the following equations:

$$r_1 + r_2 + r_3 + r_4 = 4, \tag{1.8}$$

$$r_1r_2 + r_1r_3 + r_1r_4 + r_2r_3 + r_2r_4 + r_3r_4 = 5 + \frac{b}{a}, \tag{1.9}$$

$$r_1r_2r_3 + r_1r_2r_4 + r_1r_3r_4 + r_2r_3r_4 = 4\left(\frac{b}{a}\right), \tag{1.10}$$

$$r_1r_2r_3r_4 = \frac{b}{a}. \tag{1.11}$$

Dividing (1.10) by (1.11) gives:

$$\frac{1}{r_1}+\frac{1}{r_2}+\frac{1}{r_3}+\frac{1}{r_4}=4. \tag{1.12}$$

Then, (1.8) and (1.12) imply that

$$r_1+r_2+r_3+r_4=\frac{1}{r_1}+\frac{1}{r_2}+\frac{1}{r_3}+\frac{1}{r_4}=4.$$

Now, applying the **AM-GM Inequality**, independenty, on (1.8) and (1.12) gives:

$$r_1r_2r_3r_4\leq\left(\frac{r_1+r_2+r_3+r_4}{4}\right)^4=1 \quad\Rightarrow\frac{b}{a}\leq 1 \quad\Rightarrow a\geq b. \tag{1.13}$$

$$\frac{1}{r_1r_2r_3r_4}\leq\left(\frac{\frac{1}{r_1}+\frac{1}{r_2}+\frac{1}{r_3}+\frac{1}{r_4}}{4}\right)^4=1 \quad\Rightarrow\frac{a}{b}\leq 1 \quad\Rightarrow a\leq b. \tag{1.14}$$

Thus, (1.13) and (1.14) imply that $\boxed{a=b}$, as required.

14. Mrs. Olupot gave an exam in a Mathematics class of five students. She entered the scores in random order into a spreadsheet, which recalculated the class average after each score was entered. Mrs. Olupot noticed that after each score was entered, the average was always an integer. The scores (listed in ascending order) were 71, 76, 80, 82, and 91. What was the last score Mrs. Olupot entered?

Solution:

Let $v,\ w, x\ y,\ z$ be the scores entered in the spread sheet in that order. Then the sequence of averages:

$$v,\ \frac{v+w}{2},\ \frac{v+w+x}{3},\ \frac{v+w+x+y}{4},\ \frac{v+w+x+y+z}{5},$$

are all integers. From this sequence, we observe that:

(i) v is an integer,

(ii) $v+w$ is divisible by 2,

(iii) $v+w+x$ is divisible by 3,

(iv) $v+w+x+y$ is divisible by 4, and

(v) $v+w+x+y+z=71+76+80+82+91=400$, which is divisible by 5.

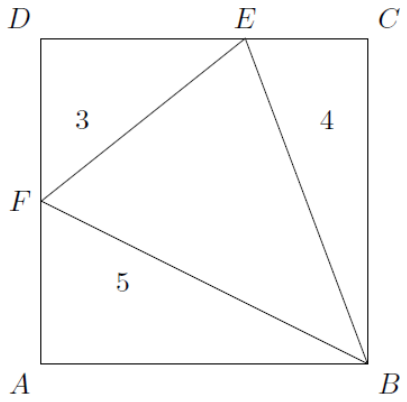
Since $v+w+x+y$ is divisible by 4 and 400 is divisible by 4, then z must also be divisible by 4. Therefore, the possible values of z are: $z=76$ or $z=80$ (the only scores divisible by 4). So, two cases arise.

- **Case 1**: For $z=76$, we have $v+w+x+y=400-76=324$. Since $v+w+x$ is divisible by 3 and 324 is divisible by 3, it follows that y must also be divisible by 3. But none of the scores, 71, 80, 82, 91 is divisible by 3. Thus, $z\neq 76$.

- **Case 2**: For $z = 80$, we have $v+w+x+y = 400-80 = 320$. Since $v+w+x$ is divisible by 3 and 320 leaves remainder 2 when divided by 3, it follows that y must also leave remainder 2 when divided by 3. Since $71 \equiv 2 \pmod 3$, $76 \equiv 1 \pmod 3$, $91 \equiv 1 \pmod 3$, we get that $y = 71$. Thus, $v+w+x = 320-71 = 249$. Since $v+w$ is divisible by 2 (or even) and 249 is odd, it follows that x must be odd. The only possibility remaining for x is $x = 91$. Thus, there are two possible cases: $(v, w, x, y, z) = (76, 82, 91, 71, 80)$ or $(v, w, x, y, z) = (82, 76, 91, 71, 80)$. In both cases $z = 80$.

Therefore, the last score that was entered in the spreadsheet is $\boxed{z = 80}$.

15. In the square $ABCD$ shown in the figure below, the areas of ΔEDF, ΔBCE and ΔFAB are 3 cm^2, 4 cm^2 and 5 cm^2, respectively. Find the area of ΔEFB.

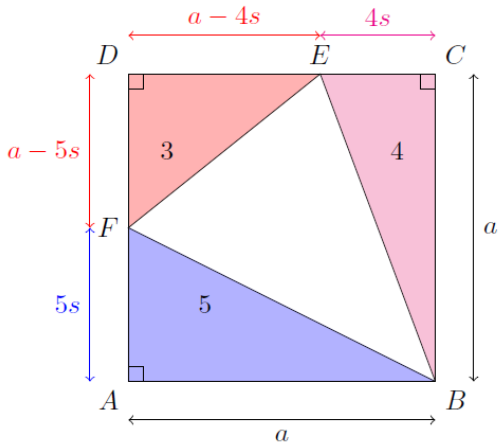


Solution:

Let a be the length of the side of the square. Considering the ratios of areas of the triangles ΔBCE and ΔFAB:

$$\frac{\text{Area of } \Delta BCE}{\text{Area of } \Delta FAB} = \frac{\frac{1}{2} \cdot a \cdot \overline{CE}}{\frac{1}{2} \cdot a \cdot \overline{AF}} = \frac{4}{5} \quad \Rightarrow \frac{\overline{CE}}{\overline{AF}} = \frac{4}{5} \quad \Rightarrow \overline{CE} : \overline{AF} = 4 : 5.$$

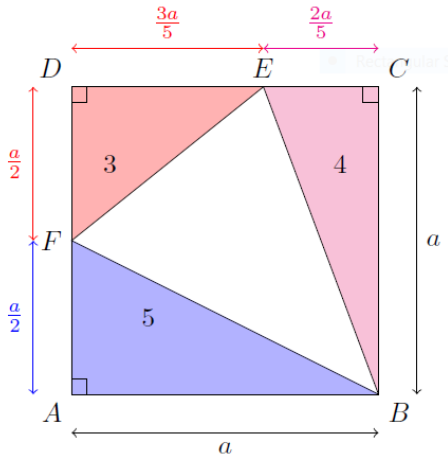
Thus, we may write $\overline{CE} = 4s$ and $\overline{AF} = 5s$ for some $s > 0$. From this, we get that $\overline{DE} = a - 4s$ and $\overline{DF} = a - 5s$. Now, consider the square below.



Next, considering the ratios of the areas of triangles ΔDEF and ΔBCE, we get that:

$$\frac{\text{Area of } \Delta DEF}{\text{Area of } \Delta BCE} = \frac{\frac{1}{2}\cdot(a-4s)\cdot(a-5s)}{\frac{1}{2}\cdot a\cdot 4s} = \frac{3}{4} \quad \Rightarrow \quad \frac{(a-4s)(a-5s)}{4as} = \frac{3}{4}.$$

This implies that $(a-4s)(a-5s) = 3as$ or $a^2 - 12as + 20s^2 = 0$ or $(a-2s)(a-10s) = 0$. Thus, either $s = \frac{a}{2}$ or $s = \frac{a}{10}$. The possibility that $a = \frac{a}{2}$ can be ruled out by observing that $5s = \overline{AF} < \overline{AD} = a$, that is, $s < \frac{a}{5}$. Using $s = \frac{a}{10}$, we can express the side lenths in terms of a only as follows: $\overline{AF} = 5s = \frac{a}{2}$ and $\overline{CE} = 4s = \frac{2a}{5}$. The new square is given in the figure below:



Considering the area of ΔFAB, we get that:

$$5 = \text{Area of } \Delta FAB = \frac{1}{2}\cdot a\cdot\frac{a}{2} \quad \Rightarrow \quad a^2 = 20 \quad \Rightarrow \quad \boxed{a = 2\sqrt{5} \text{ cm}}.$$

Thus, we get that:

$$\begin{aligned}\text{Area of } \Delta EFB &= \text{Area of } \square ABCD - (\text{Area of } \Delta EDF + \text{Area of } \Delta BCE + \text{Area of } \Delta FAB)\\ &= a^2 - (3+4+5) = 20 - 12 = 8 \text{ cm}^2.\end{aligned}$$

Therefore, $\boxed{\text{Area of } \Delta EFB = 8 \text{ cm}^2}$.